

Justification of Hardware/Software Selection

Hardware

- **HP Laptop** was for developing this application as it was convenient to work on the code & design work in the office & from home due portability. it had all the required software installed.
- **USB 2.0 8GB flash drive** was used to store the installation file & associated documentation ready for delivery.

Software

- **Microsoft Word** was used to create the design & planning documentation.
- **Google Chrome** was used for the online research & downloading Python, the associated libraries & Integrated Development Learning Environment (IDLE) which was used to code the application.
- **Python 3.12** was chosen as the compiler/interpreter because it is a procedure language, open source & free, it's easy to learn with extensive online documentation & video tutorials. Python has support for creating GUIs with libraries such as Tkinter, Custom Tkinter & JPython, allows the developer to set complex calculations & functions. Also create labels, buttons, scale sliders & combo boxes as methods for end users to input data. The code compiled with Python can be edited & extended using other compiling software like C++ making it versatile to use, it simplifies software development.
- **Auto Py To Exe** was used to convert the python script (source code) into an executable (.EXE) file so application can run independently without the need of Python & its IDLE to be installed.
- **Advance Installer** was used to create a Microsoft Software Installer (MSI) file for the EXE to simplify the installation process for the end user.